

Binary and Binomial Models (Logistic Models)

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Intended Learning Outcomes

Students will be able to:

- Fit and interpret a generalized linear model of family to binary and binomial data
- Validate these models
- Calculate the pseudo- R^2 and dispersion parameter

Recap

- Any GLM consists of three steps:
 1. Choosing a distribution for the response variable = Binary or Binomial

Binary and Binomial Data

- Can be:
 - Binary: 0,1 encoding absence/presence, survived/died
 - Binomial: probability value → 3 out of 10 survived → 0.3

$$p = \frac{k}{n} = \frac{\text{Number of successes}}{\text{number of trials}}$$



Recap

- Any GLM consists of three steps:
 1. Choosing a distribution for the response variable = Binary or Binomial
 2. Specifying the linear function of covariates and/or fixed factors

$$h(y_i) = \beta_0 + \beta_1 x_i + \varepsilon_i$$

Recap

- Any GLM consists of three steps:
 1. Choosing a distribution for the response variable = Binary or Binomial
 2. Specifying the linear function of covariates and/or fixed factors
 3. Choosing a link between the predictor function and the mean of the distribution

Logit link function

Logit Link Function

The Odds Ratio

$$\ln \left(\frac{k}{n - k} \right)$$

$\frac{\text{number of successes}}{\text{number of failures}}$

$$\ln \left(\frac{p}{1 - p} \right)$$

$\frac{\text{Probability of successes}}{\text{Probability of failures}}$

What is the odds ratio?

$$\frac{\text{number of successes}}{\text{number of failures}}$$



$$\frac{12 \text{ Alive}}{3 \text{ Dead}}$$

4 times more likely to survive

$$\frac{\text{Probability of successes}}{\text{Probability of failures}}$$



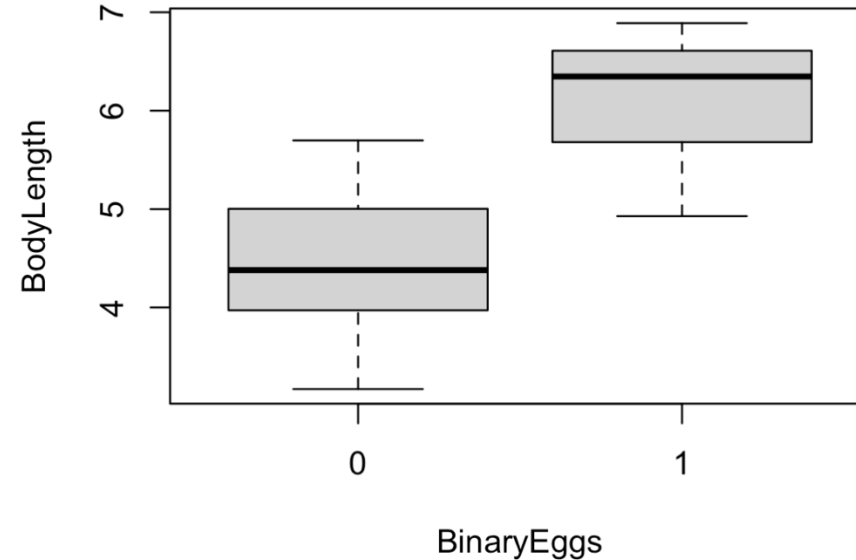
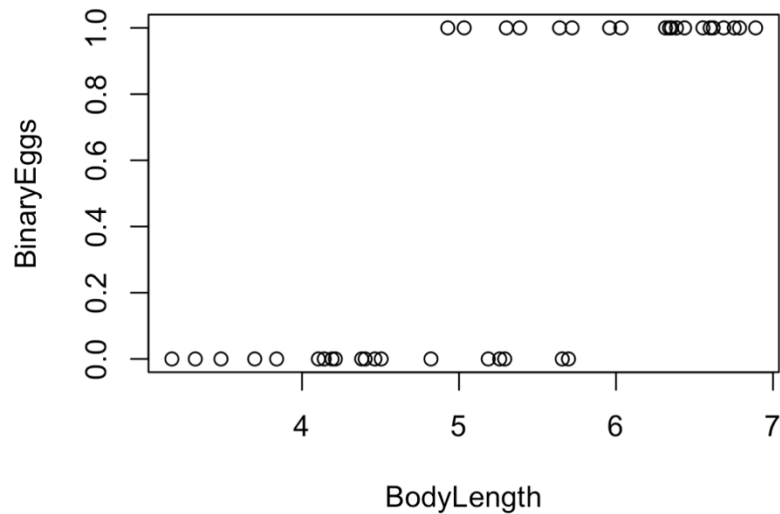
$$\frac{80\% \text{ Survival}}{20\% \text{ Survival}}$$

4 times more likely to survive

Binary Data – Moth Eggs

- Does the probability of laying an egg increase with body length?

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X \text{body length}$$



Moth Eggs Example- Binary

Call:

```
glm(formula = BinaryEggs ~ BodyLength, family = "binomial", data = motheegs)
```

Coefficients:

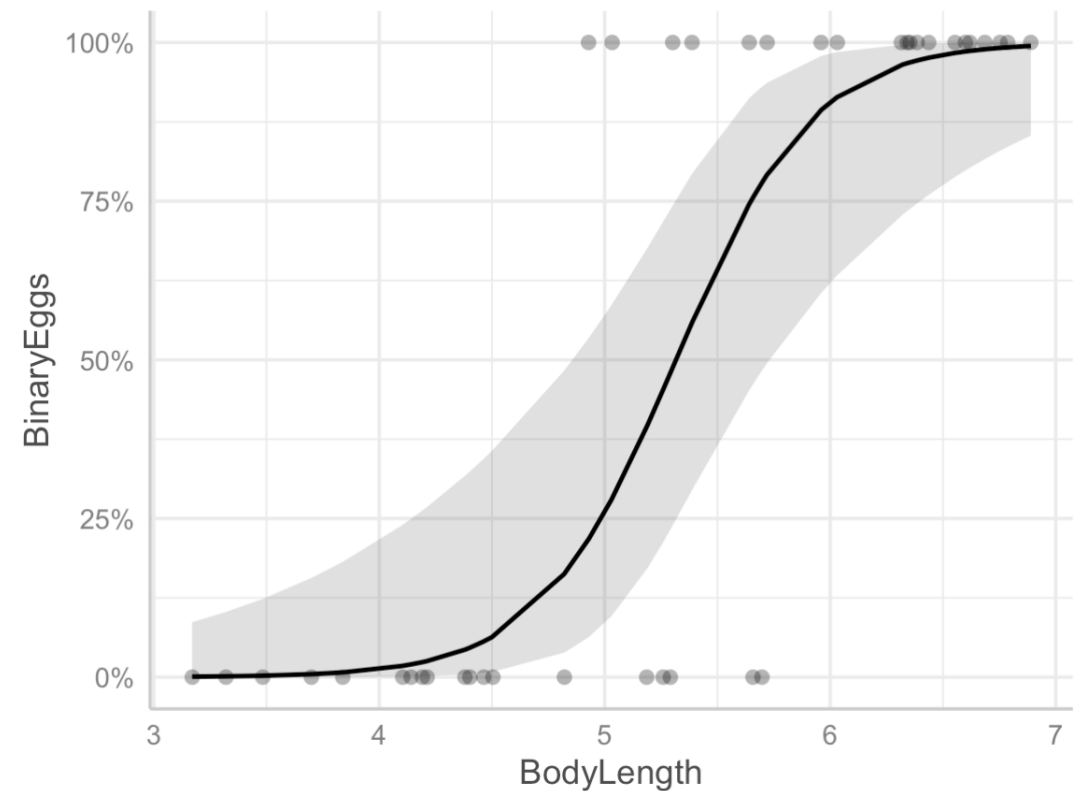
	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-17.575	5.799	-3.031	0.00244	**
BodyLength	3.306	1.079	3.064	0.00218	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 54.04 on 38 degrees of freedom
Residual deviance: 20.46 on 37 degrees of freedom
AIC: 24.46

Number of Fisher Scoring iterations: 6



Interpreting Coefficients

- The most important thing to remember with coefficients is that they are still in **the log odds ratios**.

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-17.575	5.799	-3.031	0.00244	**
BodyLength	3.306	1.079	3.064	0.00218	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- **Intercept:** Not biologically meaningful without standardization
 - Probability of laying an egg when body length is 0cm
- **Slope:** for every cm increase in body length the log odds of laying an egg increases by 3.31 or the odds of lay an egg increases by $e^{3.31} = 27.39$

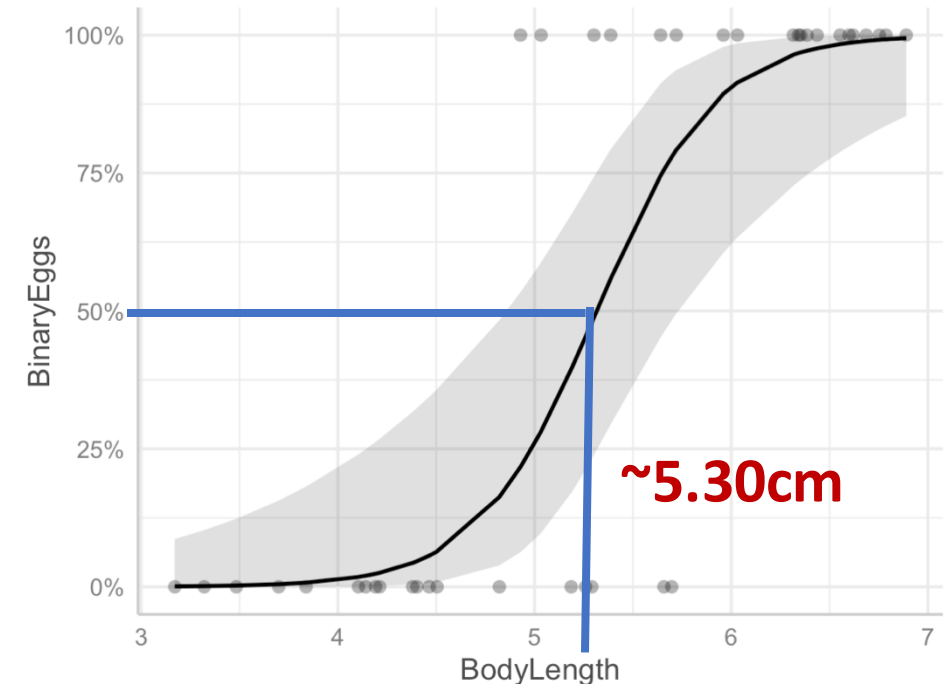
Interpreting Slope Coefficients

1. For every cm increase in body length the log odds of laying an egg increases by 3.31.
2. For every cm increase in body length the odds of laying an egg increase by 27.39.
3. Laying an egg is 27.39 times more likely for every cm increase in body length.

Finding the flip

- When thinking about probabilities we are interested in the point where the probability flips, i.e. an egg is more likely to be laid
- This occurs when the probability is greater than 0.5
- Solve for log body mass

$$0.5 = \frac{e^{-17.58+3.31Xbody\ length}}{1 + e^{-17.58+3.31Xbody\ length}}$$



Finding the flip – other solutions

- Where the values of β_0 and β_1 are absolute.

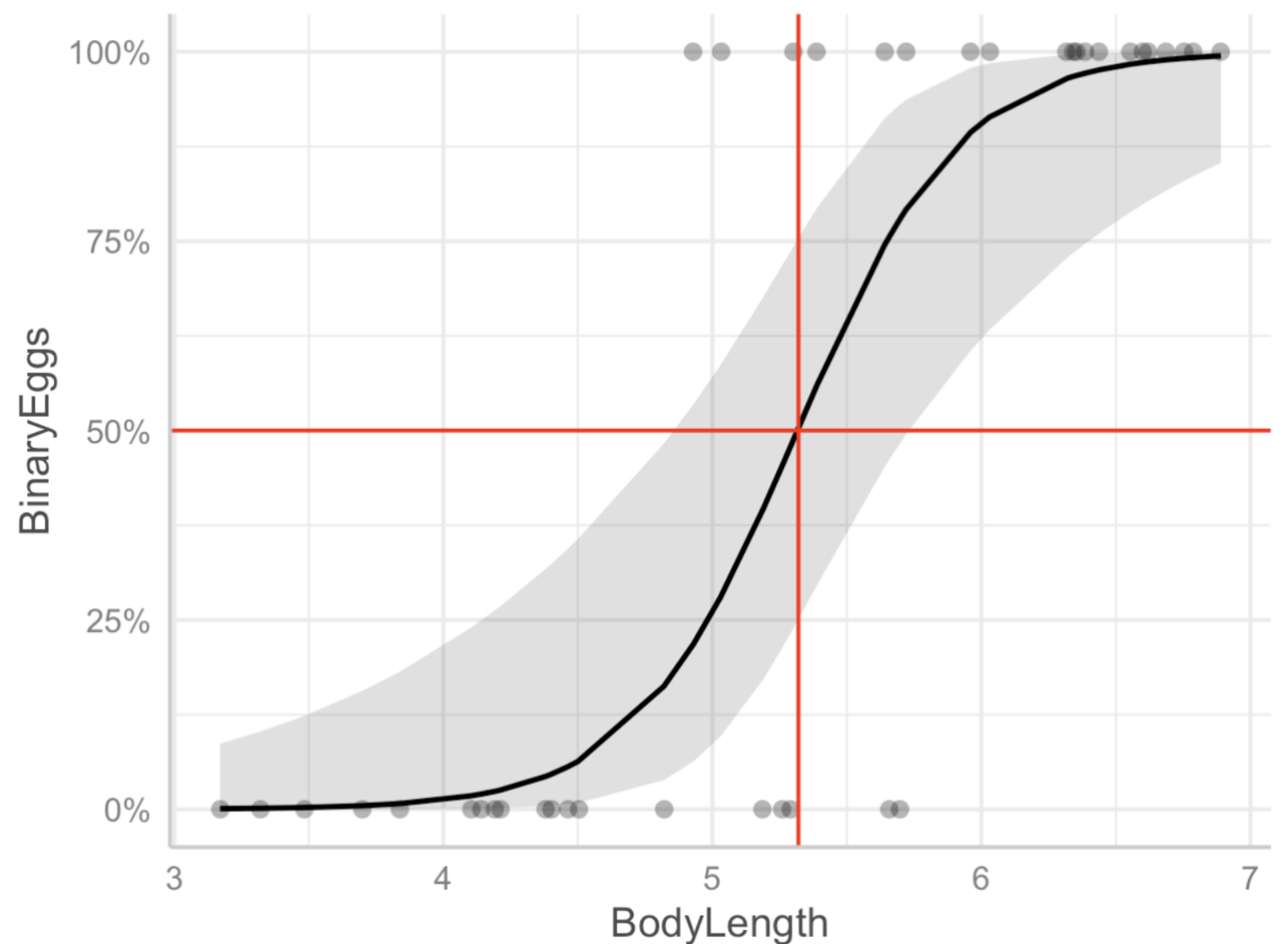
$$0.5 = \frac{e^{-17.58+3.31Xbody\ length}}{1 + e^{-17.58+3.31Xbody\ length}}$$

		Slope	
		Negative	Positive
Intercept	Negative	$-\left(\frac{-\beta_0}{-\beta_1}\right)$	$abs\left(\frac{-\beta_0}{\beta_1}\right)$
	Positive	$abs\left(\frac{\beta_0}{-\beta_1}\right)$	$-\left(\frac{\beta_0}{\beta_1}\right)$

$$abs\left(\frac{-17.58}{3.31}\right)=5.37\text{cm}$$

Finding the flip

- We can infer then that female moths with body lengths over 5.37cm are more likely to lay an egg.



Pseudo-R² and Goodness-of-fit

Call:
glm(formula = BinaryEggs ~ BodyLength, family = "binomial", data = motheegs)

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-17.575	5.799	-3.031	0.00244	**
BodyLength	3.306	1.079	3.064	0.00218	**

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(Dispersion parameter for binomial family taken to be 1)

Null deviance: 54.04 on 38 degrees of freedom
Residual deviance: 20.46 on 37 degrees of freedom
AIC: 24.46

Number of Fisher Scoring iterations: 6

62% of the variation in
egg laying probability can
be explained by female
moth body length

- Pseudo-R²:
 - $1 - (20.46 / 54.04) = 0.62$

Model Validation

- Validation is difficult as the response variable consists only of 0's and 1's and so the diagnostic plots aren't reliable.
- There is still debate regarding the use of quasi-likelihood approaches for binary data – quasi-binomial but these are regarded as controversial – they are supported by the `glm` function in R however.

Moth Eggs Example- Binomial

- **Key** assumption is that the total eggs is the same across all moths

```
Call:
glm(formula = cbind(Hatched, Eggs - Hatched) ~ BodyLength, family = "binomial",
    data = motheegs)
```

Coefficients:

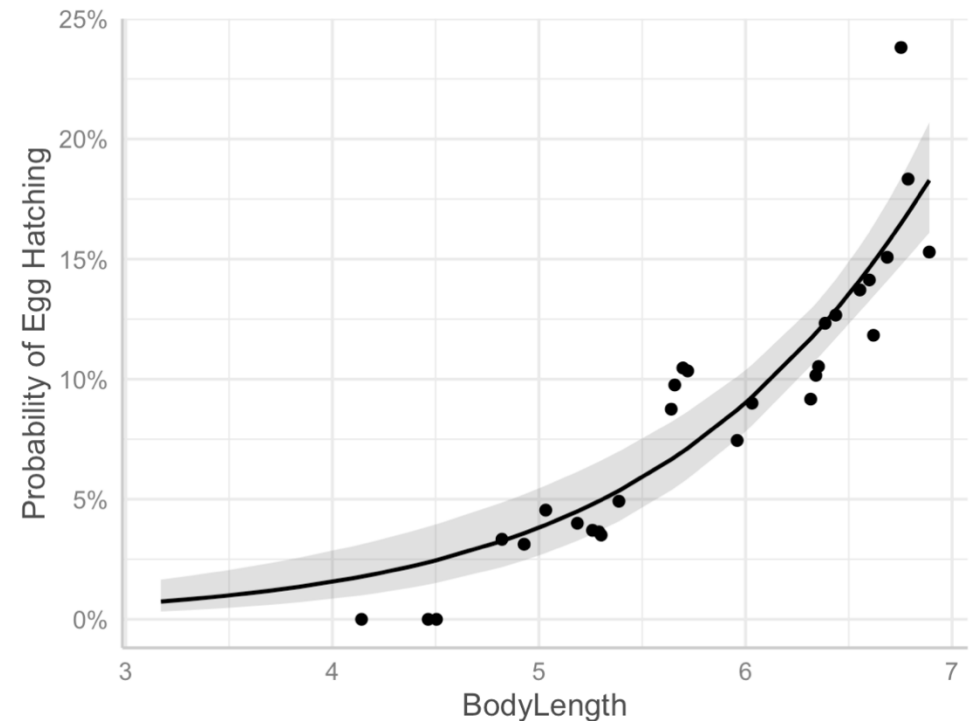
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-7.8030	0.8120	-9.609	< 2e-16 ***
BodyLength	0.9152	0.1256	7.289	3.13e-13 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 85.659 on 28 degrees of freedom
Residual deviance: 19.676 on 27 degrees of freedom
AIC: 124.62

Number of Fisher Scoring iterations: 4



Model Formula

- Does maternal body length affect the probability of eggs hatching?

```
glm(formula = cbind(Hatched, Eggs - Hatched) ~ BodyLength, family = "binomial",  
    data = motheegs)
```

The diagram illustrates the relationship between the R glm formula and the log-odds ratio. An orange arrow points from the formula `cbind(Hatched, Eggs - Hatched)` in the R code to the fraction $\frac{\text{number of eggs hatched}}{\text{number of eggs not hatched}}$. Another orange arrow points from this fraction to the log-odds ratio $\ln \left(\frac{k}{n - k} \right)$.

$$\frac{\text{number of eggs hatched}}{\text{number of eggs not hatched}} \rightarrow \ln \left(\frac{k}{n - k} \right)$$

Interpreting Coefficients

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-7.8030	0.8120	-9.609	< 2e-16	***
BodyLength	0.9152	0.1256	7.289	3.13e-13	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- For every cm increase in body length, the log odds of a moth egg hatching increases by 0.92

OR

- For every cm increase in body length, the odds of a moth egg hatching increases by $e^{0.92} = 2.5$

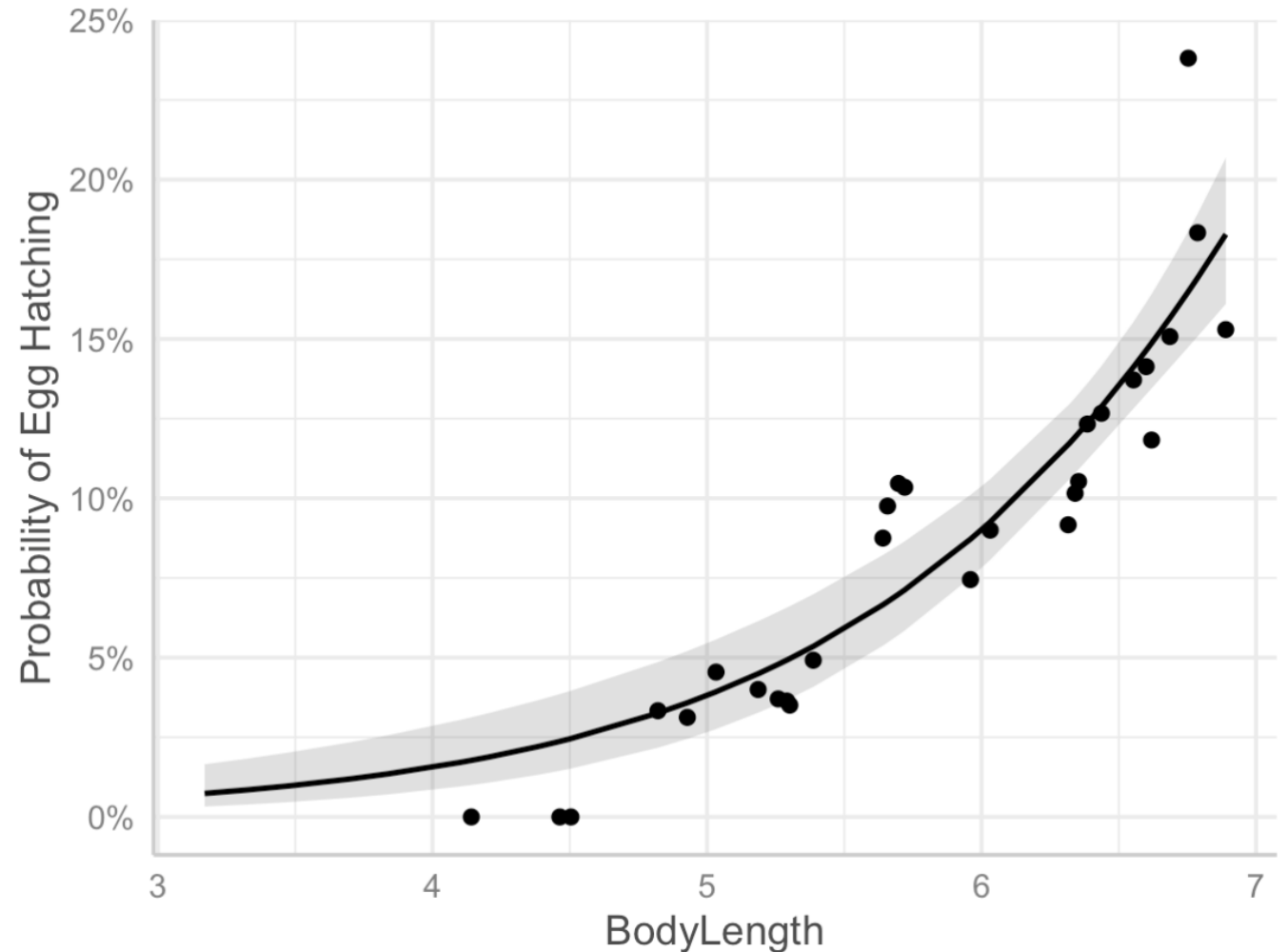
OR

- An egg hatching is 2.5 times more likely for every cm increase in body length.

We could find the flip

$$\text{abs}\left(\frac{-\beta_0}{\beta_1}\right) = \frac{-7.8030}{0.9152} = -8.53 = 8.53\text{cm}$$

- This is problematic as we extrapolating beyond our data, i.e. max moth body length is 6.8cm.



Pseudo R² and Goodness-of-Fit

Call:

```
glm(formula = cbind(Hatched, Eggs - Hatched) ~ BodyLength, family = "binomial",  
    data = motheegs)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
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(Dispersion parameter for binomial family taken to be 1)

Null deviance: 85.659 on 28 degrees of freedom
Residual deviance: 19.676 on 27 degrees of freedom
AIC: 124.62

Number of Fisher Scoring iterations: 4

77% of the variation in
egg hatching probability
can be explained by
female moth body length

■ Goodness-of-fit:

■ pseudo-R²: $1 - (19.68 / 85.66) = 0.77$

Model Validation

Null deviance: 85.659 on 28 degrees of freedom
Residual deviance: 19.676 on 27 degrees of freedom
AIC: 124.62

- $(19.676/27) = 0.73 \rightarrow$ **underdispersed**
- Underdispersion methods are less developed
- If overdispersed can fit a quasi-binomial approach and investigate the following factors:
 - Too simplistic (missing explanatory variables and/or interaction terms)
 - Explanatory variables measured on different scales
 - A covariate has a non-linear effect
 - One or more outliers
 - Zero inflation
 - Inherent dependency in the data – i.e. pseudoreplication [mixed models]

Another Example – Deer Data

- Parasite infection *Elaphostrongylus cervi* of red deer
- Measured as number of individuals infected by the parasite
- Explanatory variables:
 - Fenced (yes or no)
 - % open land, scrubs and pine plantation
 - Number of *Quercus* plants per area
- Hypothesis:
 - % open land has a negative impact on parasite burden
 - Fencing decreases parasite infection

Another Example – Deer Data

Call:

```
glm(formula = ydeer ~ OpenLand + Fenced, family = "binomial",  
     data = Tbdeer)
```

Coefficients:

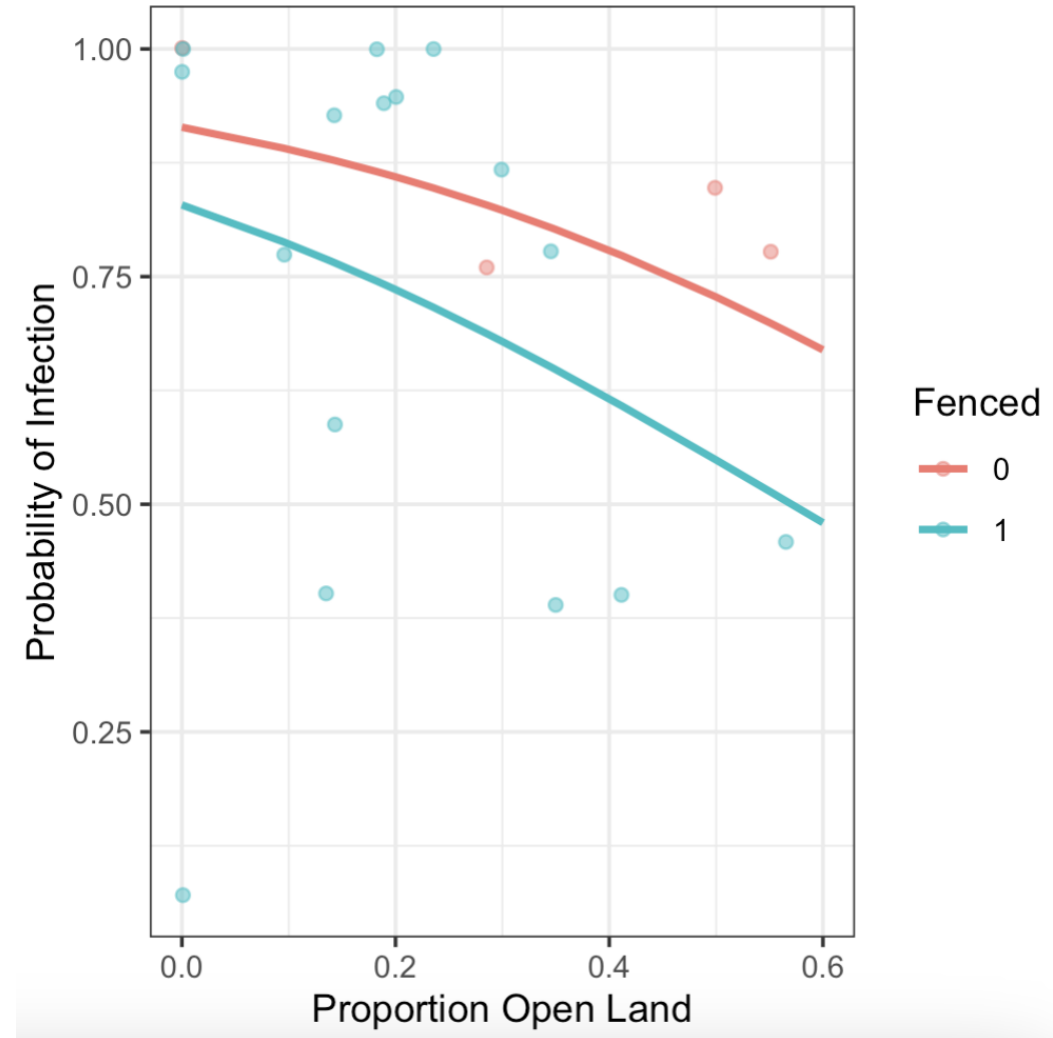
	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	2.3638	0.3558	6.644	3.05e-11	***
OpenLand	-2.7624	0.3930	-7.030	2.07e-12	***
Fenced	-0.7874	0.3394	-2.320	0.0203	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 235.58 on 23 degrees of freedom
Residual deviance: 177.74 on 21 degrees of freedom
(8 observations deleted due to missingness)
AIC: 242.82

Number of Fisher Scoring iterations: 4



Another Example – Deer Data

- What are the model equations for:

- Unfenced:

$$\ln\left(\frac{p}{1-p}\right) = 2.37 - 2.76 * OpenLand$$

- Fenced:

$$\ln\left(\frac{p}{1-p}\right) = 2.37 - 2.76 * OpenLand - 0.79$$

$$\ln\left(\frac{p}{1-p}\right) = 1.58 - 2.76 * OpenLand$$

Call:

```
glm(formula = ydeer ~ OpenLand + Fenced, family = "binomial",  
     data = Tbdeer)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	2.3638	0.3558	6.644	3.05e-11	***
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Another Example – Deer Data

- What are the flip points for:

Call:

```
glm(formula = ydeer ~ OpenLand + Fenced, family = "binomial",  
     data = Tbdeer)
```

- Unfenced:

$$\text{abs}\left(\frac{2.37}{-2.76}\right) = 0.86$$

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	2.3638	0.3558	6.644	3.05e-11	***
OpenLand	-2.7624	0.3930	-7.030	2.07e-12	***
Fenced	-0.7874	0.3394	-2.320	0.0203	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- Fenced:

$$\text{abs}\left(\frac{1.58}{-2.76}\right) = 0.57$$

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Residual deviance: 177.74 on 21 degrees of freedom
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Number of Fisher Scoring iterations: 4

Another Example – Deer Data

- What are the model equations for:

- Unfenced:

$$\ln\left(\frac{p}{1-p}\right) = 2.37 - 2.76 * OpenLand$$

- Fenced:

$$\ln\left(\frac{p}{1-p}\right) = 2.37 - 2.76 * OpenLand - 0.79$$

$$\ln\left(\frac{p}{1-p}\right) = 1.58 - 2.76 * OpenLand$$

- R-squared: $1 - \left(\frac{177.74}{235.58}\right) = 0.25$

- Dispersion parameter: $177.74/21 = 8.46$

Call:

```
glm(formula = ydeer ~ OpenLand + Fenced, family = "binomial",  
     data = Tbdeer)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	2.3638	0.3558	6.644	3.05e-11	***
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AIC: 242.82

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Summary

- Logistic models can handle binary outcomes (0,1) and binomials via the log odds ratio
- The logit function abstracts the interpretation of the coefficients and instead plotting is preferable
- Logistic models fitted to binary data are hard to validate and quasi-likelihood approaches are generally avoided, however, these can be implemented with binomial data
- All generalized linear models can include random effects to address to non-independence and/or correct for overdispersion